

SKILL1

1)Installation

Compiling and executing a C program

```
aasritha@DESKTOP-4B676UM:~$ mkdir first
aasritha@DESKTOP-4B676UM:~$ cd first
aasritha@DESKTOP-4B676UM:~/first$ nano hello.c
aasritha@DESKTOP-4B676UM:~/first$ gcc hello.c -o hello
aasritha@DESKTOP-4B676UM:~/first$ ./hello
Hello, Linux!
aasritha@DESKTOP-4B676UM:~/first$ |
```

2)Linux Shell and Basic Commands

a) pwd

```
aasritha@DESKTOP-4B676UM:~$ pwd
/home/aasritha
```

b) ls (List Files and Folders)

```
aasritha@DESKTOP-4B676UM:~$ ls
OS  OSLab  cd  hello  hello.c  shortcut.txt
aasritha@DESKTOP-4B676UM:~$
```

c) cd (Change Directory)

```
aasritha@DESKTOP-4B676UM:~/OSLab$ mkdir Lab
aasritha@DESKTOP-4B676UM:~/OSLab$ cd Lab
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$
```

d) mkdir (Make Directory)

```
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ mkdir OSLab
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ ls
OSLab
```

e) rmdir (Remove Empty Directory)

```
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ mkdir Project
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ ls
OSLab Project
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ rmdir Project
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ ls
OSLab
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ |
```

f) cp (Copy Files)

```
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ echo "Hello Linux" > file1.txt
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ cp file1.txt file2.txt
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ ls
OSLab file1.txt file2.txt
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ |
```

g) mv (Move or Rename)

```
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ mv file2.txt newfile.txt
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ ls
OSLab file1.txt newfile.txt
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$
```

h) rm (Remove File)

```
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ rm newfile.txt
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ ls
OSLab file1.txt
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ |
```

i) cat (Display File Content)

```
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ cat file1.txt
Hello Linux
```

j) echo (Print Text)

```
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ echo "Welcome to Linux"
Welcome to Linux
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ echo "Operating Systems Lab" > lab.txt
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ cat lab.txt
Operating Systems Lab
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$
```

k) find (Search for Files)

```

operating systems Lab
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ find .
.
./lab.txt
./OSLab
./file1.txt
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ find . -name "lab.txt"
./lab.txt

```

I) history (View Previous Commands)

```

aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ history
 1  sudo apt update && sudo apt upgrade -y          # Step 6
 2  sudo apt install gcc -y                          # Step 7
 3  gcc --version                                     # Step 8 - verify
 4  ls
 5  gcc hello.c -o hello
 6  ./hello
 7  pwd
 8  ls
 9  uname
10  whoami
11  uname -a
12  sudo apt update
13  sudo apt install gcc -y
14  gcc --version
15  nano hello.c
16  ls
17  gcc hello.c -o hello
18  ./hello
19  pwd
20  lscpu
21  free -h
22  lsblk
23  ps
24  top
25  echo $SHELL

```

m) shell usage

The **shell** is the program that accepts your commands and executes them.

```

aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ pwd
/home/aasritha/OSLab/Lab
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ ls
OSLab  file1.txt  lab.txt
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ mkdir Test
aasritha@DESKTOP-4B676UM:~/OSLab/Lab$ cd Test
aasritha@DESKTOP-4B676UM:~/OSLab/Lab/Test$

```

Git Repository and Project Structure - Git installation, git init, git add, git commit, git status, project directory organization (src, include, bin, docs)

```
aasritha@DESKTOP-4B676UM:~$ mkdir -p ~/projects/os-lab
cd ~/projects/os-lab
aasritha@DESKTOP-4B676UM:~/projects/os-lab$ cd ~/projects/os-lab
git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/aasritha/projects/os-lab/.git/
aasritha@DESKTOP-4B676UM:~/projects/os-lab$
```

```
aasritha@DESKTOP-4B676UM:~/projects/os-lab$ git config --global user.name "Aasritha"
git config --global user.email "aasritha@example.com"
aasritha@DESKTOP-4B676UM:~/projects/os-lab$ git config --list
user.name=Aasritha
user.email=aasritha@example.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
aasritha@DESKTOP-4B676UM:~/projects/os-lab$ git add .
git commit -m "Initial project setup"
git status
On branch master

Initial commit

nothing to commit (create/copy files and use "git add" to track)
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

```
compilation terminated.
aasritha@DESKTOP-4B676UM:~/projects/os-lab$ nano src/hello.c
aasritha@DESKTOP-4B676UM:~/projects/os-lab$ gcc src/hello.c -I include -o bin/hello
aasritha@DESKTOP-4B676UM:~/projects/os-lab$ ./bin/hello
Hello from src!
```

Shell Architecture and Makefile - User → Shell → Kernel → Hardware, Bash, process execution, writing a simple Makefile, using make

```
aasritha@DESKTOP-4B676UM:~$ mkdir Skill
aasritha@DESKTOP-4B676UM:~$ cd Skill
aasritha@DESKTOP-4B676UM:~/Skill$ nano hello.c
aasritha@DESKTOP-4B676UM:~/Skill$ nano Makefile
aasritha@DESKTOP-4B676UM:~/Skill$ gcc hello.c -o hello
aasritha@DESKTOP-4B676UM:~/Skill$ ./hello
Hello, Linux!
aasritha@DESKTOP-4B676UM:~/Skill$ ls
Makefile  hello  hello.c
```

```
aasritha@DESKTOP-4B676UM:~/Skill$ make
Makefile:14: *** missing separator.  Stop.
aasritha@DESKTOP-4B676UM:~/Skill$ |
```

Process Creation using fork() - Process abstraction, fork(), parent and child processes, getpid(), getppid(), multiple fork() examples

CODE:

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
int main() {
```

```
    pid_t
```

```
    pid = fork();
```

```
    if (pid < 0) {
```

```
        perror("fork failed");
```

```
    } else if (pid == 0) {
```

```
        printf("Child: my PID is %d\n", getpid());
```

```
    } else {
```

```
        printf("Parent: my child's PID is %d\n", pid);
```

```
    }
```

```
return 0;
```

```
}
```

```
aasritha@DESKTOP-4B676UM:~$ nano forkdemo
aasritha@DESKTOP-4B676UM:~$ nano forkdemo.c
aasritha@DESKTOP-4B676UM:~$ gcc src/forkdemo.c -o bin/forkdemo
./bin/forkdemo
```

```
Parent: my child's PID is 800
Child: my PID is 800
```